Climate Investment and Employment Creation:

A Case Study of Morocco

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Abstract

Employment co-benefits are often used to support the case for climate investment, but their size and timing depend on how investment flows into labor demand. This paper estimates domestic labor demand associated with climate investment and related co-benefits along a net-zero pathway for Morocco. It links cost-benefit outputs from the greenhouse gas emissions model SISEPUEDE (Kalra et al., 2023) to MINDSET, a multi-regional input-output framework (Dorbrand et al., 2025). We compare baseline and low-emissions scenarios over 33 periods and estimate employment associated with the gradual implementation of 18 climate policies, or transformations, in Morocco's decarbonization portfolio. The low-emissions pathway supports positive net employment gains over time, amounting to roughly 602,000 job-years and stabilizing near 37,000 jobs in later periods. Two transformations with strong links to construction activities stand out for their job creation potential: appliance efficiency and building insulation improvements. These findings show how the pipeline can distinguish not only the scale but also the employment content of climate investment pathways over time, providing evidence to better align climate and development agendas.

1. Introduction

Governments pursuing deep emissions reductions also need to consider how the transition will affect domestic employment. This question is especially important in developing economies with rapidly growing working-age populations and high levels of informal employment. Decarbonization and employment growth are often presented as complementary, partly because low-carbon activity may be more labor-intensive than the fossil-fuel activity it displaces (Hanna et al., 2024). The International Labour Organization (ILO) estimates that the energy transition could create 24 million jobs and eliminate 6 million by 2030 (Knudsen et al., 2025; World Bank, 2025). Whether this pattern holds within a particular country and over the timetable of a net-zero pathway depends on what the transition requires, where the relevant goods and services are produced, and how labor-intensive the domestic industries supplying them are.

Existing estimates answer this question only partly. The evidence is concentrated in rich countries. Of the 121 studies in the most recent systematic review, six concern Africa, mostly South Africa, while the fourteen that report employment per dollar invested draw only on European, North American, and Indian data (Hanna et al., 2024). Most estimates also provide a single horizon or regional aggregate, offering limited evidence on when employment effects emerge. Montt et al. (2018), for example, report a single 2030 snapshot. Detailed occupational evidence remains specific to particular settings and should not be readily extrapolated (Bücker et al., 2025). Many aggregate estimates also allow workers to move from declining sectors into growing ones without fully accounting for the frictions that may constrain this reallocation (Knudsen et al., 2025).

This paper estimates the net direct and indirect domestic labor demand associated with a low-emissions pathway for Morocco. We compare two scenario runs from SISEPUEDE, a greenhouse gas emissions model (Kalra et al., 2023), which generates a calibrated decarbonized scenario for Morocco as a case study. We compare between a baseline pathway and a low-emissions pathway calibrated to reach net-zero emissions. For each climate action, or transformation, we construct annual flows of investments or avoided spending and calculate the employment estimates through MINDSET, a multi-regional input-output framework developed by the World Bank that draws on the 120 products of the GLORIA database[1] (Dorbrand et al., 2025). Morocco's input-output structure, domestic supply shares, and employment coefficients then convert the changes in final demand into estimates of domestic labor demand by sector and period over a 33-period horizon.

The estimates do not show an immediate or uniformly positive labor-demand effect relative to the baseline. The estimated net change is negative from t_3 through t_8, and later levels off at the equivalent of roughly 37,000 jobs above the baseline. Roughly 85% of the cumulative net labor-demand gain accrues from t_{19} onward. Two transformations account for over 90% of the net total. Reducing building heat demand through improved building insulation supports 290,577 job-years, while raising industrial energy efficiency supports 242,443 job-years. Their large employment totals reflect not only the scale of spending but also the sectors in which that spending is allocated.

The rest of the paper proceeds as follows. Section 2 reviews the employment literature. Section 3 presents the data and estimation. Section 4 describes Morocco's economic and climate context and presents the main results. Section 5 discusses what the estimates can and cannot support, and Section 6 concludes.

[1] GLORIA (Global Resource Input-Output Assessment) database is a multi-regional input-output (MRIO) covering 163 regions and 120 products, including bilateral trade flows. https://ielab.info/labs/ielab-gloria

2. Literature Review

The employment effects of green investment, comparing the economic advantages of green spending against other investments, have received substantial attention in the literature. Systematic reviews comparing higher-carbon and low-carbon activities find that the fossil fuel sector creates, on average, 5 jobs per USD million invested, compared with a range of 15 to 22 jobs per USD million for low-carbons industries, such as renewable energy and energy efficiency technologies (Hanna et al., 2024; O'Callaghan et at, 2022). Employment estimates have been developed along three main methodological lines, namely Input-Output (IO), computable general equilibrium (CGE), and macro-econometric models, each has particular strengths and limitations.

Input-output (I-O) frameworks, including national and multi-region input-output (MRIO) models[2], have been increasingly used to estimate job creation potential in the low-carbon transition (Bücker et al., 2025; San Miguel Arrieta et al., 2024; Černý et al., 2021; Montt et al., 2018). These frameworks trace how a dollar of exogenous final demand propagates through inter-industry supply chains. They translate this demand into sectoral gross output and intermediate input requirements, capturing direct and indirect job creation at fine sectoral and regional resolution. Several studies in this tradition assume fixed cost breakdowns when constructing demand vectors (Bücker et al., 2025; Salman et al., 2025; Bakry et al., 2023; Ram et al., 2020; Garrett-Peltier, 2017)[3]. Yet the underlying expenditure structure may differ across countries and may change as technologies mature, which limits the extent to which these models can trace a particular intervention to jobs in a specific setting. Employment multipliers may be misleading when they do not account for variation in capital-to-labor composition, the import shares of major components, the labor intensity of production, and prevailing wage levels (Agan and Balcilar, 2022; Perrier and Quirion, 2018). The same concern applies in dynamic settings, where capital and operational expenditure trajectories depend on the initial expenditure structure.

Computable general equilibrium (CGE) models go further by allowing prices, wages, and factor substitution to adjust while imposing optimization and market clearing. Some extensions produce net employment estimates that account for displacement and imperfect labor mobility (Mealy et al., 2025; Castellanos and Heutel, 2024; Fragkos & Fragkiadakis, 2022; Chateau et al., 2018). However, investments in green technology and mitigation actions are often represented using fixed cost structures that may not fully capture the underlying economic dynamics. As CGE results are sensitive to production function calibration and closure assumptions, inaccurate capital-labor shares introduced in the models may distort production function estimates and factor substitution patterns, biasing simulated labor-demand responses (Castellanos and Heutel, 2024; Bolarinwa, 2024; Hafstead et al., 2022). When the inputs of green technologies and climate investments do not reflect country-specific import dependence and domestic supply capacity, models may allocate demand across domestic industries incorrectly and generate imprecise job gains. For solar photovoltaic deployment, for example, domestic employment depends partly on the

[2] Grounded in national accounts and official statistics, MRIO databases (e.g., GLORIA, Eora) are increasingly used for empirical analysis as they provide a time-series of high resolution on interindustry and trade linkages across global supply chains, in addition to environmental and socio-economic extensions.

[3] Because current IO tables do not explicitly identify many green technologies or investment activities, such as renewable energy industries (e.g., solar, bioenergy), methodological extensions typically estimate employment multipliers by constructing weighted averages of existing industries represented in the I-O tables (Garret-Peltier, 2017; Pollin, 2016). Weights and industries involved usually rely on fixed cost structures from survey data and the climate-policy literature.

extent to which panels and electrical equipment are imported rather than produced locally. This may shift labor demand away from manufacturing and toward installation, project development, maintenance, and other local services.

A third tradition uses macro-econometric models, including energy-environment-economy (3E) frameworks such as E3ME and NEMESIS, to estimate how key economic variables (e.g., GDP, employment) adjust over time using historical data. These models have been applied particularly to the economic analysis of European energy and climate policy (Alexandri et al., 2024; Ortega et al., 2020; Lehr and Lutz, 2019; Pollitt et al., 2015). These models are demand-led in the short to medium term and combine national accounting identities with estimated behavioral equations. These equations assess how demand shocks – policy shocks, new investment – affect sectoral output, which then feeds into the labor demand through sectoral production functions and wage-setting equations, avoiding the market clearing assumptions of standard CGE models (Cambridge Econometrics, 2022; Akcigit et al., 2022). However, their core behavioral relationships are estimated from historical time series, embedding sectoral interdependencies observed in the past, which may be less reliable for forward-looking policy analysis. This is particularly relevant where the cost and input structures of the sectors being modelled are changing rapidly. In these cases, historical coefficients may not represent the employment implications of a particular intervention, such as climate investments, in a specific country context.

Across these approaches, fixed cost breakdowns, cost trajectories derived from an initial structure, and projections based on historical relationships may not fully capture variation in the conditions that shape employment across countries. This issue may become more important where climate policy requires substantial changes in technologies, production practices, and supply chains rather than incremental adjustments. Employment outcomes are also commonly reported as aggregate sectoral effects. Fewer studies link a portfolio of climate interventions, specified at particular implementation levels, to employment outcomes in a country-specific pathway toward net zero.

3. Data and Methods

3.1. Data and analytical framework

This paper estimates net direct and indirect domestic labor demand by sector over a 33-year period[4]. The estimates are based on climate investments linked to a policy portfolio of 18 transformations designed to support a country's transition to net-zero emissions[5]. The portfolio draws on simulation outputs from SISEPUEDE[6], a greenhouse gas emissions model (Kalra et al., 2023) that defines sector-level climate actions and simulates decarbonization scenarios given a country's economic structure and local characteristics. Climate investments are modeled as annual purchases deployed through investment flows over the analysis period, with expenditures determined by the cost structures of the green technologies and low-carbon activities introduced over time. For the employment analysis, we use MINDSET, a multi-

[4] Direct jobs are created in the industry producing the final output, while indirect jobs arise in upstream industries that supply the inputs needed for production.

[5] In total, SISEPUEDE's climate action portfolio includes 41 transformations, of which 18 are included in the employment estimates.

[6] SISEPUEDE (Simulation of Sectoral Pathways and Uncertainty Exploration for Decarbonization) is a toolbox for designing development strategies and analyzing, under uncertainty, their emissions, costs, benefits, and alignment with development goals and Paris Agreement targets (Kalra et al., 2023).

regional input-output framework developed by the World Bank that draws on the GLORIA database[7] (Dorbrand et al., 2025). Morocco is used as our case study to examine how transformation cost structures translate into employment estimates within a particular economic structure.

SISEPUEDE produces two scenario runs for Morocco over the analysis period. The baseline scenario follows a business-as-usual path (BAU) consistent with historical development trajectories,[8] including modest improvements in the transition to renewables and energy efficiency, while fossil fuels remain in use throughout the economy. The low-emissions pathway implements a portfolio of 18 transformations across sectors at gradually increasing levels of implementation calibrated to achieve net-zero emissions.[9] For each transformation, we take the difference between the two runs in the SISEPUEDE output variable that tracks its physical effect, such as industrial energy consumption or emissions avoided, and multiply that difference by a unit cost to build an annual investment flow, reported in constant USD. These flows define the amount and timing of spending or savings over the analysis period.

We use transformation-specific cost structures, drawn from the climate literature, to allocate each monetary flow across the sectors that supply the relevant goods and services. For each expenditure or savings line, the cost structure provides a sectoral composition that is used to construct a demand vector, defined as a weighted bundle of products in the input-output table. The weights are calculated as the relative share of each cost component mapped to its corresponding producing sector and sum to one (Miller and Blair, 2022; Garrett-Peltier, 2017). The shares are used to allocate the monetary flow across GLORIA's 120 products, which we aggregate to 20 sectors for reporting.[10] Savings enter as negative final demand and are assigned to the supply-chain industries associated with the fossil fuels that are no longer purchased, rather than to the industries receiving the investment. Electrifying light-duty vehicles, for example, produces one line for vehicle and charging-equipment purchases and a separate negative line for fuel that is no longer purchased. The latter withdraws demand from products linked to petroleum, chemicals, and transport activities. In short, SISEPUEDE generates the monetary flows, while the cost structures determine their sectoral allocation.

Employment factors, including the labor share of an industry's output, and inter-industry linkages are drawn from GLORIA, which uses 2019 as its base year and includes bilateral trade flows (Lenzen et al., 2022). In our case study, MINDSET applies Morocco's 2019 trade patterns, domestic production shares, and sector-specific employment coefficients when translating demand into jobs. Figure 1 provides an overview of the pipeline, showing the quantity carried at each step and the separate paths that expenditure and savings follow before rejoining in the employment step.

In principle, cost structures could vary with deployment scale as the composition of expenditure changes with technological maturity. Early-stage climate investments may be capital-intensive and concentrated in equipment and construction, often with high import content (Bücker et al., 2025). As

[7] GLORIA (Global Resource Input-Output Assessment) database is a multi-regional input-output (MRIO) covering 163 regions and 120 products, including bilateral trade flows. https://ielab.info/labs/ielab-gloria

[8] In Kalra et al. (2023), a "Traditional Development" trajectory follows past development trends associated with rising GDP per capita and growing demand for goods and services (e.g., beef consumption, automobile ownership).

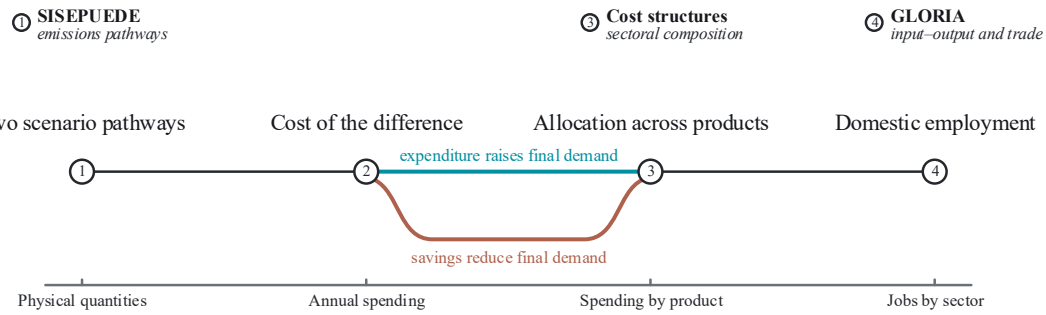
[9] SISEPUEDE integrates the modeling of agriculture, forestry, and other land use (AFOLU), energy, transport, buildings, industry, and waste, following the methodology of the Intergovernmental Panel on Climate Change (IPCC, 2006, 2019).

[10] GLORIA's sector classification follows the United Nations International Standard Industrial Classification of All Economic Activities (ISIC) Revision 4 (United Nations Statistics Division, 2008).

deployment expands, operational expenditure may account for a larger share of total spending and direct more demand toward domestically supplied maintenance and services. In the present implementation, time variation in cost structures arises only where a transformation depends on the annual electricity-generation mix. The mix changes over time as the scenario converges toward a generation profile consistent with net-zero emissions.

MINDSET translates cost structures into employment estimates under a demand-led model that assumes spare capacity and involuntary unemployment (Dorbrand et al., 2025). Under this assumption, demand shocks generate additional output without crowding out other activity or requiring factor reallocation.[11] The calculation combines the Leontief inverse and employment coefficients derived from GLORIA with trade shares computed from its final-demand tables. In the input-output framework, the Leontief inverse traces the direct and indirect output required throughout supply chains to meet a given change in final demand (Miller and Blair, 2022). Trade shares then determine the portion of product demand supplied domestically rather than imported and, in turn, the portion that can generate domestic employment. Employment coefficients then translate the resulting output changes into direct and indirect jobs. The estimates use this demand-driven quantity and employment channel, with prices held at their base-year values. They are therefore conditional on transformation-level cost structures, Morocco's input-output structure, trade shares, and employment factors. Figure 1 summarizes the pipeline, showing the quantity carried at each step and the separate paths that expenditure and savings follow before they rejoin in the employment estimates.

Figure 1. Overview of the estimation pipeline



3.2.Employment multipliers

To compute employment effects, we use GLORIA at its 120-product resolution. Rather than include an explicit “Rest of the World” block that consolidates the database's remaining 162 regions, we account for imports at two points. First, only the share of final demand for each product that is met by domestic producers enters the calculation. Second, the Leontief inverse – $(I_{120\times120} - A_{120\times120})^{-1}$ – traces the supply-chain effects of this demand through the multi-regional system, while employment is retained only for domestic output. Equation 1 gives the change in domestic output associated with a change in final demand. Equation 2 converts that output change into employment by pre-multiplying it with an employment-output matrix, which provides the number of workers per unit of output in each industry,

[11] These conditions may be more plausible in the short and medium run than market clearing with full employment, especially in developing countries such as Morocco, where labor-market slack and informality are substantial. That said, we agree that these assumptions may be less plausible in the long run.

producing the employment-requirement (ER) table. Direct employment captures the jobs in the industry supplying the product purchased, and indirect employment covers the jobs generated along its supply chain (Equation 3). Because demand vectors are separated by transformation and year, employment can be reported by landing sector and as annual totals over the 33-year period, as shown in Equation 4.

In these expressions, $\Delta \mathrm{y}_{j,t}$ is the demand vector for transformation j in year t, built from its cost structure and expressed in thousands of USD. A is the technology matrix for domestic production, which measures how much input from one sector is needed to produce one unit of output in another sector. $(\mathrm{I}-\mathrm{A})^{-1}$ is the resulting Leontief inverse. $\hat{\mathrm{s}}$ and $\hat{\mathrm{e}}$ are diagonal matrices of domestic supply shares and employment coefficients, respectively. The element $ER_{i,p}$ gives domestic jobs in product i per thousand USD of final demand booked to product p. ER^{dir} is captures direct employment. Subscripts j and t index transformations and years in the analysis period, respectively. Subscript k indexes the reporting sectors. N_t is the number of jobs supported in year t. For Morocco, the calculation applies the country's 2019 domestic supply shares, production structure, and product-specific employment coefficients when translating demand into jobs. All input-output procedures, including matrix extraction and employment calculations, are documented in Python and R and are available in the Supplementary Materials.

$$\Delta \mathrm{x}_{j,t} = (I_{120\times120} - A_{120\times120})^{-1} \cdot \hat{\mathrm{s}} \cdot \Delta \mathrm{y}_{j,t} \qquad (1)$$

$$\Delta \mathrm{n}_{j,t} = ER_{120\times120} \cdot \Delta \mathrm{y}_{j,t}, \qquad ER = \hat{\mathrm{e}} \cdot (I - A)^{-1} \cdot \hat{\mathrm{s}} \qquad (2)$$

$$\Delta \mathrm{n}_{j,t}^{dir} = \hat{\mathrm{e}} \cdot \hat{\mathrm{s}} \cdot \Delta \mathrm{y}_{j,t}, \qquad \Delta \mathrm{n}_{j,t}^{ind} = \Delta \mathrm{n}_{j,t} - \Delta \mathrm{n}_{j,t}^{dir} \qquad (3)$$

$$N_{k,t} = \sum_j \sum_{i \in S_k} \Delta n_{i,j,t}, \qquad N_t = \sum_k N_{k,t} \qquad (4)$$

Because each flow is computed from a change in a physical variable, its sign follows that change. A decline in fuel consumption, for example, enters as a negative value. We then apply an economic orientation rule. Expenditure lines increase final demand and support employment, while savings lines reduce it by withdrawing spending that previously supported employment in fossil-fuel supply chains. Net employment is the sum of these positive and negative effects. One limitation of this approach is that savings are not reallocated within the model. Avoided expenditure therefore reduces demand without replacement. The estimates exclude employment that might be supported by redirected household consumption, for example. The net figure is thus a partial accounting of final-demand changes.

4. Results

4.1. Case Study: Morocco

Morocco, a lower-middle-income country in northwestern Africa, faces substantial climate vulnerability. Temperatures have increased faster than the global average since the 1960s. At the same time, the country has an energy-intensive economy that remains highly dependent on fossil fuels, which

underlines the need for a sustained transition toward lower-carbon energy sources.[12] Morocco is also among the most water-stressed countries globally.[13] Rainfed agriculture (e.g., wheat and barley production, accounts for 80 percent of cultivated land, employs most of the agricultural workforce, and is particularly exposed to changes in water availability and crop yields (World Bank, 2022). Water scarcity could reduce GDP by up to 6.5% over the long term and contribute to urban out-migration of up to 1.9 million people (5.4% of the population) by 2050 (World Bank, 2022). Between 2015 and 2024, rural employment fell by close to one quarter (1.2 million workers). This period also saw increasingly frequent and severe droughts. Taken together, these conditions make climate action important for limiting Morocco's exposure to worsening climate risks. Climate investments that generate employment may offer one way to pursue development objectives while addressing longer-term climate needs.

Morocco's greenhouse gas emissions have risen substantially over the past two decades.[14] In 2018, the energy sector accounted for 65.1% of total gross emissions, followed by agriculture at 22.1% (World Bank, 2022). Reliance on imported fossil fuels[15], and the resulting exposure to international price shocks, have contributed to the expansion of renewable energy in the country, which accounted for around 20% of the power generation mix in 2021 (World Bank, 2022). As rainfall has declined, hydropower's contribution to installed capacity has fallen, while solar and wind generation have expanded in later years, drawing on Morocco's solar and wind resources. Coal-fired capacity nevertheless accounted for 39% of total generation capacity in 2021 and remained a major source of emissions in the power sector (World Bank, 2022).

Agriculture is highly exposed to climate conditions and accounts for 13% of GDP, employing 26.3% of Morocco's workforce in 2023 (World Bank, 2026). Between 2000 and 2023, the share of workers in agriculture fell (-19.6 p.p.), shifting workers to industry and services (World Bank, 2026). Rural labor-market employment has followed an overall downward trend, with substantial fluctuations when drought reduces rainfed crop production (World Bank, 2022). Morocco is the world's second-largest phosphate producer and holds approximately 70% of known global phosphate reserves (U.S. Geological Survey, 2026). The state-owned OCP Group has plans to increase clean-energy use, including through green ammonia. Green ammonia[16] may be used in long-distance transport and as a substitute for natural gas in power generation (Adeli et al., 2023; World Bank, 2022). Morocco's automotive industry has become an important part of its manufacturing and export sectors. The country has attracted electric-vehicle investment as European automakers expand production. Its close economic links with the European Union may support continued access to European markets and further investment in lower-carbon manufacturing (World Bank, 2026; World Bank, 2022). These conditions shape the economic and employment context for climate mitigation investments in Morocco.

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[12] In 2019, fossil fuels accounted for 90.2 percent of Morocco's total energy use, including oil at 56.5 percent, coal at 29.8 percent, and natural gas at 3.9 percent. Although renewable energy accounted for around 20 percent of electricity generation by 2021, Morocco's energy system remained strongly dependent on fossil fuels (World Bank, 2022).

[13] Between 1960 and 2020, Morocco's per capita renewable water availability declined from about 2,560 m^3 per person per year to roughly 620 m^3. This level is below the 1,000 m^3 threshold for structural water stress and close to the 500 m^3 threshold for absolute water scarcity (World Bank, 2022).

[14] Between 2000 and 2019, Morocco's greenhouse gas emissions (GHG) doubled from 44.6 to 91.2 million tons of CO2 equivalent (World Bank, 2022).

[15] Between 2010 and 2020, energy accounted for 19.4% of Morocco's total imports.

[16] Green ammonia (NH_3) can be produced using renewable electricity to generate hydrogen through water electrolysis, avoiding the CO_2 emissions associated with conventional ammonia production from natural gas. (Bouaboula et al., 2023).

4.2. Main Results

Our estimates cover the 18 transformations for which cost structures are defined. Together they involve $60.7 billion of additional spending relative to the baseline and $16.7 billion of avoided spending, summed across the 33-year period covered in this exercise. Net employment is below the baseline from t_3 to t_8 and above it from t_9 onward. By t_{33}, net employment is 37,166 jobs above the baseline, approximately 0.3% of Morocco's labor force[17]. The sum of annual net employment effects over the 33 periods is 601,737 job-years, nearly 85% of which accrue from t_{19} onward. A job-year denotes one job sustained for one period. The total therefore reflects labor demand over the horizon rather than the number of individuals employed[18]. Figure 2 shows the annual series by sector, with net employment across all sectors shown as a line.

The timing of the result follows the timing and composition of spending. Employment coefficients are fixed over time, while spending depends on the level of implementation. Implementation begins for most of the portfolio around t_9 and approaches its maximum late in the horizon. As implementation approaches its maximum, employment from direct investment purchases, grows more slowly, while employment associated with displaced spending continues to rise as the transformations' co-benefits are realized (e.g., avoided fuel costs). The series flattens as these effects offset each other. The 601,737 net job-years are the difference between 785,532 job-years on the investment side and 183,795 job-years on the displaced side.

Two transformations account for 88.6% of the net total. Reducing building heat demand through improved building insulation supports 290,577 job-years, while raising industrial energy efficiency supports 242,443 job-years. Their large employment totals reflect not only the scale of spending but also the sectors in which that spending is allocated. Both are tied to the same reduction in building energy use, but appliance efficiency is assumed to cost twice as much per unit saved, which helps explain the difference in total spending. Appliance efficiency accounts for $22 billion and returns 3.3 job-years per million dollars. Improved building insulation accounts for $11 billion and returns 26.4 job-years per million dollars. Building insulation work places nearly 50% of its employment in construction, which has relatively high employment multipliers compared to other sectors[19] (Ernst and Sarabia, 2015; Kofoworola and Gheewala, 2008).

Sector figures show where employment effects are recorded rather than which transformations generate them. Each sector gains demand from some transformations and loses it from others, and the reported total is the difference between these flows. Construction accounts for 206,078 job-years, about a third of the total. This concentration comes mainly from building-shell work, rather than from a general tendency for low-emissions spending to land in construction. Wholesale and retail trade receives little direct expenditure, but records 103,570 job-years because distribution margins are applied to activity in other sectors. Across the portfolio, 355,744 job-years are direct and 245,993 are indirect. Indirect effects account for about 41% of total employment as these jobs arise in domestic supply chains rather than in the industry that receives the spending.

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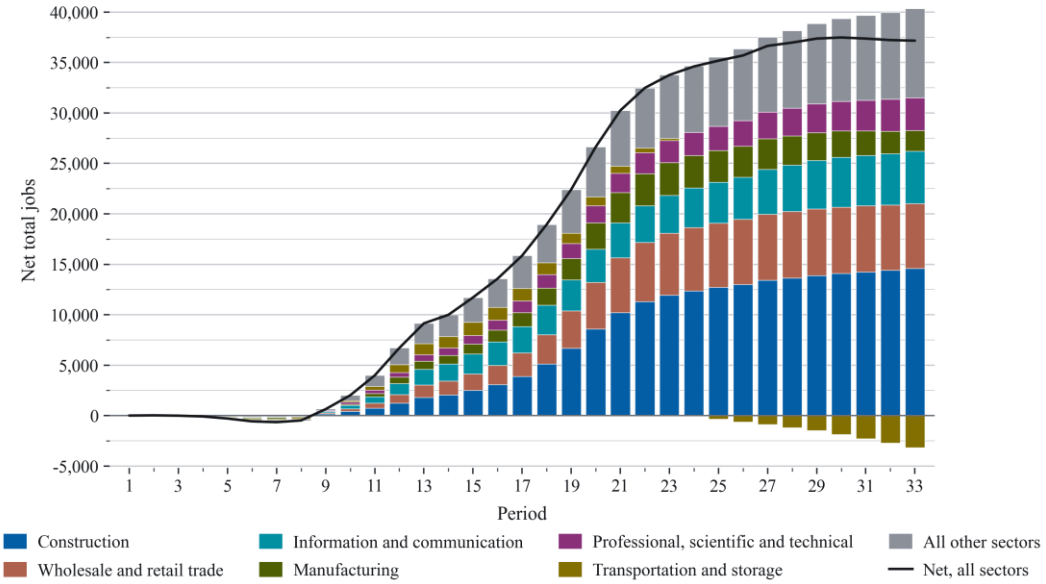
[17] In 2025, Morocco labor force size reached 12.6 million workers (World Bank Data, 2026). https://data.worldbank.org/indicator/SL.TLF.TOTL.IN?locations=MA

[18] The measure captures the volume of employment supported over time, not the number of distinct workers. A position that lasts ten periods contributes ten job-years, for instance.

[19] Construction sector multipliers can exceed those for the economy as a whole because of characteristics of construction activity, such as the reliance on material inputs from other industries and relatively labor-intensive work. (Ernst and Sarabia, 2015; Kofoworola and Gheewala, 2008)

Transportation and storage is the only sector with a negative cumulative employment effect, at 3,203 job-years over the full period. Investment in other transformations generates 65,005 job-years in the sector because transport services are used in the activities represented by those investments. At the same time, avoided expenditure reduces employment in transportation and storage by 68,208 job-years. Fuel no longer purchased for light-duty vehicles accounts for 78 percent of this reduction. The timing of these two effects determines the sector's net result. Displaced fuel first reaches the sector at t_{16}. Until then, transportation and storage is a net gainer, peaking at 1,307 jobs at t_{15}. By t_{33}, employment in the sector is 3,168 jobs below the baseline. These estimates capture employment changes associated with final demand. The estimates do not show how this adjustment would occur within the sector.

Figure 2. Net employment by sector across analysis period



5. Discussion

Using Morocco as a case study, the analysis shows that a low-carbon pathway can shift modeled labor demand unevenly over time and across investment lines. The estimates point to a modest aggregate change in labor demand, but the aggregate figure alone is not enough to assess the adjustment implied by the pathway. The timing of demand and its distribution across investment lines are also relevant. Modeled labor demand increases gradually over the transition, rather than peaking sharply over a short period. This may avoid concentrating changes in labor demand within a narrow window, but it also means that the increase emerges later in the transition. Evidence from other settings indicates that workers do not necessarily move quickly from contracting to expanding activities, particularly when related occupations decline together (Bücker et al., 2025; Donovan and Schoellman, 2023).

The variation in employment response across sectors where climate investments are made also matters. Hanna et al. (2024) identify technological maturity, local content, the labor-capital split, and prevailing wages as possible sources of this variation. Montt et al. (2018) likewise find larger employment changes in construction than in manufacturing because construction is more labor-intensive. Project design, procurement, and local sourcing may therefore affect the amount and timing of modeled labor demand associated with a given level of spending. This does not mean that investments should be selected to

maximize jobs per unit of spending. Higher labor intensity may reflect lower productivity rather than a social gain (Hanna et al., 2024).

The analysis models labor demand rather than realized employment. It does not observe whether workers can move into expanding activities, whether firms can fill the jobs implied by the pathway, or how employment changes would be distributed across workers and regions. Knudsen et al. (2025) note that net-positive employment estimates commonly assume relatively efficient movement from declining to growing sectors, which is not as often observed, particularly in developing settings. The authors identify the role of training, active labor-market programs, and social protection as important for translating potential jobs into realized employment. The available evidence provides only partial information on these conditions in Morocco. Informal employment in Northern Africa affects over 60% of workers, beside a labor-force participation of 43% (ILO, 2026). In Morocco, clean energy firms report technical-skills shortages as an obstacle to decarbonization-driven job growth, and women remain underrepresented in construction and energy, where modeled labor demand is concentrated (World Bank, 2025). These indicators do not identify the share of modeled demand that would become realized employment. The results therefore identify the timing of modeled labor demand and the investment lines associated with larger demand effects, rather than the pathway's effects on realized employment, labor-force exit, earnings, or welfare. If employment losses occur before labor demand expands, income support may need to reach workers who are not recorded as unemployed. Measures intended to reduce work disruption or help workers enter expanding activities address a different problem.

6. Conclusion

This paper estimates domestic labor demand associated with climate investment and related co-benefits along a net-zero pathway for Morocco. It links cost-benefit outputs from the greenhouse gas emissions model SISEPUEDE (Kalra et al., 2023) to MINDSET, a multi-regional input-output framework (Dorbrand et al., 2025). We compare baseline and low-emissions scenarios over 33 periods and estimate employment associated with the gradual implementation of 18 climate policies, or transformations, in Morocco's decarbonization portfolio.

Relative to the baseline, estimated net labor demand is negative over the first eight periods and positive thereafter. It reaches a peak of 37,481 jobs above the baseline before ending at 37,166 jobs in the final period. The annual effects sum to 601,737 job-years over the 33-year period. This timing is consistent with the pathway's expenditure profile, which front-loads capital investment and has later growth in operation and maintenance spending and savings associated with the transformations (Bücker et al., 2025). The estimates identify employment associated with changes in final demand over time.

Sectoral results report where employment effects are recorded, rather than which transformations generate the underlying demand. Construction accounts for 206,078 job-years, about one-third of the total, largely because of improved building insulation work. Wholesale and retail trade records 103,570 job-years through distribution margins on activity in other sectors. Direct effects account for 355,744 job-years and indirect effects for 245,993. Indirect effects therefore account for 41% of estimated employment through domestic supply chains. Transportation and storage is the only sector with a negative cumulative effect, at 3,203 job-years. It receives 65,005 job-years associated with investment but loses 68,208 job-years from avoided expenditure, mainly fossil fuel no longer purchased for light-duty vehicles.

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